INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

STARTERS, ENGINE, ELECTRICAL: 24-VOLT DIRECT-CURRENT

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 24-volt (V) direct-current (dc) starters for cranking nonaircraft internal-combustion engines (see 6.1).

1.2 Classification. Starters shall be of the following types, grades, classes, and duties, as specified (see 6.2):

Type I	- For direct (engine crankshaft) engagement.
Type II	- For peripheral (ring gear) engagement.
Grade A	- Waterproof.
Grade B	- Nonwaterproof.
Class 1	- Positive drive - solenoid.
Class 2	- Inertia drive.
Class 3	- Positive drive - mechanical.
Duty	- Light (see 6.4.2).
Duty	- Heavy (see 6.4.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2920

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-S-571	- Solder, Tin Alloy: Tin-Lead Alloy; and Lead Alloy.

MILITARY

MIL-P-514	- Plates; Identification, Instruction and Marking, Blank.
MIL-F-13927	- Fungus Resistance Test; Automotive Components.

STANDARDS

FEDERAL

FED-STD-H28/2	- Screw - Thread Standards for Federal Services Section 2 Uniform Inch Screw Threads - UN and UNR Thread Forms.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1184	- Electrical Components for Automotive Vehicles; Waterproofness Tests.
MIL-STD-2073-1	- DoD Material Procedures for Development and Application of Packaging Requirements.

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MS53008	- Starter, Engine, Electrical, 24-Volt, D.C., 5-9/16-Inch Frame Diameter, Solenoid-Actuated, Heavy-Duty.
MS53011	- Starter, Engine, Electrical, 24-Volt, D.C., 5-1/8-Inch Frame Diameter, Solenoid-Actuated, Heavy-Duty, Long-Frame.
MS53013	- Starter, Engine, Electrical, 24-Volt, D.C., 3-Inch Frame Diameter, Solenoid-Actuated, Light-Duty.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

7338603	- Brush Assembly.
8737873	- Brush Assembly.
8737877	- Solenoid.
11602684	- Drive Assembly.
11667957	- Drive Assembly.
11667959	- Drive Assembly.

(Copies of drawings required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The starters furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) at the time of award of contract (see 4.4 and 6.3).

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3.1.1 Authorized repair parts. Repair parts furnished under this specification shall be only those specified for authorized repair (see table I), and shall be products which are qualified (see 3.1.1.1 and 3.1.1.2) for listing on the applicable QPL at the time set for opening of bids.

3.1.1.1 Assembled repair parts. Repair parts requiring qualification to this specification will not be automatically qualified when those parts have been contained in a starter assembly that has been qualified, unless they pass the specific tests for those repair parts (see 4.4.1).

3.1.1.2 Separate repair parts. Repair parts submitted separately for qualification shall be qualified as follows: Repair parts shall be assembled with all parts necessary to form a complete starter assembly. A repair part shall be considered qualified when the starter assembly containing that repair part has met the repair part qualification requirements of this specification (see 4.4.1).

TABLE I. Authorized repair parts.

Drawing number	CAGE 1/	Description
11602684	19207	Drive assembly
11667959	19207	Drive assembly
11667957	19207	Drive assembly
7338603	19207	Brush assembly
8737873	19207	Brush assembly
8737877	19207	Solenoid
71035	35510	Brush kit
71220	35510	Solenoid kit
MFJ-2012DS	19728	Brush set
5702643	19728	Brush kit

1/ Commercial and Government Entity (CAGE).

3.2 Materials. Materials shall be as specified herein, on the applicable standard or drawing (see 3.3.10), and in referenced specifications and standards (see 4.8.1).

3.2.1 Solder. Solder employed in making electrical connections in control circuits shall conform to QQ-S-571. A rosin flux shall be used (see 4.8.1).

3.2.1.1 Soldering. Soldering shall conform to MIL-STD-454, requirement 5 (see 4.8.1).

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3.2.2 Recycled, virgin, and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.4.3).

3.3 Design and construction. The starter shall be designed and constructed to the form and dimensions specified on the applicable drawing or military standard (see 3.3.10), and as specified herein (see 4.8.2).

3.3.1 Use of dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which would foster galvanic action, should be avoided. Where such contact is not necessary to complete an electrical circuit, but is otherwise unavoidable, parts shall be insulated. Dissimilar metals are defined in MIL-STD-889 (see 4.8.1).

3.3.2 Standard parts. Military standard parts shall be incorporated wherever possible and interchangeability shall be maintained for all repair parts specified on applicable drawings. Commercial standard parts may be used provided they are interchangeable with military standard parts (see 4.8.1).

3.3.3 Threaded parts. Screw threads of the form, number per inch, and class specified on the applicable drawing or military standard shall conform to FED-STD-H28/2 (see 4.8.1).

3.3.4 Locking devices. Lockwashers, self-locking nuts, safety wires, or other approved locking devices shall be incorporated where specified or required to prevent loosening of components (see 4.8.2).

3.3.5 Drive (see 4.8.1).

3.3.5.1 Type I. The drive shall consist of a starter (meshing) jaw, a torque-limiting clutch, and an automatic engaging and disengaging mechanism for meshing with the engine crankshaft drive.

3.3.5.1.1 Clutch. The torque-limiting clutch shall be adjusted to slip under the torque load specified (see 3.4.1.1).

3.3.5.1.2 Starter (meshing) jaw. The jaw shall conform to the applicable drawing or military standard and shall be designed for specified rotation (see 3.3.5.3).

3.3.5.2 Type II. The drive shall consist of a pinion for meshing with the flywheel ring gear and a means for effecting such meshing. The drive shall be designed for specified rotation (see 3.3.5.3). Seals shall be provided as specified on the applicable drawing or military standard.

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3.3.5.2.1 Class 1. The pinion shall employ a positive drive for meshing with the flywheel ring gear. The pinion shall be actuated into mesh by a solenoid.

3.3.5.2.2 Class 2. The pinion shall employ an inertia drive for meshing with the flywheel ring gear. De-meshing shall occur within specified revolutions per minute (rpm) limits.

3.3.5.2.3 Class 3. The pinion shall be actuated into mesh by mechanical means and shall employ a positive drive for meshing with the flywheel ring gear.

3.3.5.3 Rotation. Unless otherwise specified (see 6.2), the direction of the rotation of the drive, as viewed from the driving end of the starter, shall be clockwise.

3.3.6 Mounting flange. The mounting flange shall conform to the applicable drawing or military standard (see 4.8.1).

3.3.7 Terminals. All terminal sizes and locations shall be in accordance with the applicable drawing or military standard (see 4.8.1).

3.3.8 Brushes. Brushes shall be readily accessible for inspection and replacement. Brush pigtails shall be prevented from contacting conducting parts having a difference of potential with respect to the pigtails (see 4.8.1).

3.3.9 Interchangeability. All solenoid assemblies and authorized repair parts shall be physically and functionally interchangeable with all other products of the same Part or Identifying Number (PIN) (see 4.8.2).

3.3.10 Drawings (see 4.8.1).

3.3.10.1 Military standards and drawings. The applicable military standard (MS53008, MS53011, MS53013, or other) or drawing shall be as specified by acquisition activity (see 6.2).

3.3.10.2 Contractor drawings. All drawings pertaining to the starter including performance curve, assemblies, sub-assemblies, and details, shall be maintained by the contractor. Unless otherwise specified on the drawing or military standard, the performance curve shall be plotted based on the current heavy-duty Society of Automotive Engineers (SAE) battery curve for heavy-duty motors and the current standard SAE battery curve for light-duty motors, and shall conform to the SAE standard form for starting motor characteristic curves, except that it shall be optional to plot the horsepower curve beyond 80% of peak horsepower (negative slope) (see 6.2).

3.4 Operating requirements.

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3.4.1 Performance. When tested as specified in 4.8.3 through 4.8.6, the starter shall evidence no damage, and shall conform to the performance curve as specified on the applicable drawing or military standard.

3.4.1.1 Type I. The starter shall meet the following performance requirements:

a.	Voltage (at starter terminals)	13.5 ± 0.5 V.
b.	Jaw speed (minimum)	25 rpm.
c.	Continuous load	400 pound-foot (lb-ft).
d.	Efficiency (minimum)	65 percent (%).
e.	Slipping torque	850 ± 100 lb-ft.

3.4.2 Position (type II only). The starter shall operate within specified performance requirements (see 3.4.1) and without evidence of damage when the longitudinal axis is in any angular position within 45 degrees of the intended operating position (see 4.8.18.2.1 and 4.8.18.3).

3.4.3 Overspeed capability. When tested as specified in 4.8.7, the starter shall evidence no damage. After the test, the starter shall meet the requirements of 3.4.1.

3.4.4 Dielectric strength. When tested as specified in 4.8.8, the insulation shall evidence no burning, charring, loosening or cracking (arcing), smoking, rupture, or other damage. After the test, the starter shall meet the requirements of 3.4.1.

3.4.5 Loaded static torque. When tested as specified in 4.8.9, pinion and clutch dentils shall have consistent engagement without hesitation or loss of applied torque up to 350 lb-ft.

3.4.6 Tooth abutment load. When tested as specified in 4.8.10, on indexing drives, a maximum force of 38 pounds, when applied axially to a single pinion tooth at its pitch diameter shall move the pinion to its full indexing position.

3.4.7 Overrunning torque. When tested as specified in 4.8.11, pinion and clutch dentils on the heavy duty positork type clutch shall be separated between 10,000 and 14,000 rpm. When pinion and clutch dentils are separated, the overrunning torque shall be less than 8 pound-inch at 15,000 rpm.

3.5 Environmental requirements.

3.5.1 Waterproofness, grade A.

3.5.1.1 Qualification and control tests. When tested as specified in 4.8.12.1, the starter shall evidence no leakage that would adversely affect performance. The starter shall meet the requirements of 3.4.1 subsequent to the submersion test.

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3.5.1.2 Quality conformance tests. When tested as specified in 4.8.12.2, the starter shall evidence no bubbles indicating internal leakage.

3.5.2 Corrosion resistance. When tested as specified in 4.8.13, the starter shall evidence no corrosion affecting operation and shall subsequently meet the requirements of 3.5.1, 3.4.4, and 3.4.1.

3.5.3 Fungus resistance. After having been subjected to the test specified in 4.8.14, the starter shall evidence no microbial growth that would adversely affect performance. Subsequent to the test, the starter shall meet the requirements of 3.4.1 and 3.4.4.

3.5.4 Vibration resistance. When tested as specified in 4.8.15, components of the starter shall not be loosened or damaged. The starter shall subsequently meet the requirements of 3.4.1.

3.5.5 Shock resistance. When tested as specified in 4.8.16, components of the starter shall not be loosened or damaged. The starter shall subsequently meet the requirements of 3.4.1.

3.6 Life cycle requirements.

3.6.1 Life test (live engine test). When tested as specified in 4.8.17, drives shall have a minimum life of 45,000 cycles. Three consecutive start attempt failures because of tooth abutment (failure of drive pinion to engage ring gear) shall be defined as a drive failure. In addition, failure of the drive to transmit adequate torque to the ring gear shall be defined as a failure.

3.6.2 Endurance. When tested as specified in 4.8.18, there shall be no need for servicing or replacement of parts. Following the test, each starter shall operate within 10% of the performance previously established by the requirements as specified in 3.4.1.

3.6.2.1 Classes 1 and 2. When tested as specified in 4.8.18, no malfunctions of the solenoid, its components, or engaging linkages shall occur. Failure of the drive to overcome abutments shall not exceed 7% of the total cycles accumulated at any stage of the test. The drive pinion shall not mill the flywheel or mating gear used.

3.6.2.2 Type I. When tested as specified in 4.8.18.1, the starter shall complete the 6,400 cycles without replacement of, or damage to any of the component parts. The total brush wear shall not have exceeded 80% of the original effective brush length on any one brush. (Original effective brush length equals the distance from the end of the brush to the 100% wear line. This equals 0.265 inch for brushes conforming to Drawing 7338603.) After the test, the starter efficiency shall not have decreased more than 10% from the original value.

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3.6.2.3 Brush wear (Classes 1 and 2). When tested as specified in 4.8.18.2.3, total brush wear shall not exceed 50% of the original effective brush length on any one brush. Pinpoint sparking is allowed. Excessive or irregular sparking on one or more brushes shall be considered brush failure.

3.6.2.4 Extreme temperature. When tested as specified in 4.8.18.4, the drive and clutch assemblies shall operate satisfactorily in applicable cranking motors at -65 degrees Fahrenheit (°F) and +160°F. During the test there shall be free movement on the armature shaft, and pinion-to-clutch dentils shall function properly.

3.7 Painting. Exterior surfaces of the starter shall be painted with a black gloss enamel (see 4.8.1 and 4.8.2).

3.8 Marking (see 4.8.2).

3.8.1 Permanent mark. The military PIN shall be permanently inscribed on the frame in such a location as not to be covered by the nameplate. Where a military PIN has not been assigned, this marking may be omitted (see 6.2).

3.8.2 Nameplate. Nameplate materials shall conform to MIL-P-514 and marking to MIL-STD-130. The size of the nameplate shall be proportioned to the starting motor. Legibility of marking shall be adequate for normal 20/20 vision and shall be accomplished in a manner satisfactory to the Government. The nameplate shall be firmly attached to the exterior of the starter frame in the manner and position indicated on the applicable drawing or military standard and such that accidental loss or loosening of the nameplate will not affect environmental characteristics or performance of the starter. The nameplate shall include the following minimum information:

Starter, 24 V dc
Rotation (indicated by arrow)
Specification MIL-S-3785
Military PIN (see 6.2)
Federal stock number (see 6.2)
Manufacturer's serial number
Manufacturer's name or trademark.

3.9 Workmanship. Workmanship shall be such as to assure a product free of rust, burrs, scratches, chips, sharp edges, loose or defective terminals, cracked insulation, or faulty soldering (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase

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order (see 6.2), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).
b. Quality conformance inspection (QCI) (see 4.5).
c. Control tests (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed under the following standard (room ambient) conditions:

a. Temperature: 77 $\pm$ 18°F.
b. Relative humidity: Uncontrolled room ambient.
c. Atmospheric pressure: Site pressure.

4.3.1 Temperature stabilization. Unless otherwise specified on the applicable drawing or military standard, the starter shall be stabilized for 1 hour at 77 $\pm$ 18°F before being subjected to testing.

4.3.2 Power supply and resultant horsepower. The power supply shall be capable of reproducing a battery curve that conforms to the SAE standard-duty curve or the SAE 24-V heavy-duty curve which ever is applicable. Unless otherwise specified on the applicable drawing or military standard, a variation of 10% in resultant horsepower as indicated on the applicable drawing or military standard shall be allowed for each sample submitted for test.

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4.4 Qualification inspection (starters). A qualification sample of four starters of each type, grade, class, duty, and PIN for which qualification is being requested shall be furnished for qualification testing. One additional sample (total five) shall be furnished for type II, heavy-duty starters, sample E only. The samples shall be representative of the units proposed to be furnished to the Government. Qualification inspection shall be conducted by the Government at a place designated by the Government and shall consist of inspection as specified in 4.8.1; examination for the defects specified in table II; and testing as specified in table III, in the order listed (see 6.3.1).

TABLE II. Classification of defects.

Category	Defects	Method of inspection
Major		
101	Dimensions affecting interchangeability, out of tolerance (see 3.3 and 3.3.9).	SIE 1/
102	Faulty workmanship affecting performance (see 3.9).	Visual
Minor		
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3).	SIE
202	Locking of threaded parts, improper (see 3.3.4).	Visual
203	Painting, improper (see 3.7).	Visual
204	Marking, improper (see 3.8).	Visual
205	Faulty workmanship affecting appearance (see 3.9).	Visual

1/ Standard inspection equipment (SIE).

TABLE III. Order of qualification testing.

Sample	Paragraph	Test
A	4.8.3 thru 4.8.6	Operational
	4.8.18.1 thru 4.8.18.3	Endurance
	4.8.3 thru 4.8.6	Operational

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TABLE III. Order of qualification testing - Continued.

Sample	Paragraph	Test
B	4.8.3 thru 4.8.6	Operational
	4.8.7	Overspeed capability
	4.8.12.1	Waterproofness
	4.8.8	Dielectric strength
C	4.8.3 thru 4.8.6	Operational
	4.8.15	Vibration resistance
	4.8.14	Fungus resistance
D	4.8.3 thru 4.8.6	Operational
	4.8.16	Shock resistance
	4.8.13	Corrosion resistance
E (duty-heavy only)	4.8.3 thru 4.8.6	Operational
	4.8.18.4.1	Endurance (extreme temperature)
	4.8.3 thru 4.8.6	Operational

4.4.1 Qualification inspection (drives, brushes, and solenoids). Qualification samples shall consist of the following quantities of authorized repair parts:

Part	Number of samples
Drive assembly 11602684, 11667957, 11667959	5
Brush assembly 7338603	8
Brush assembly kit 71035	1
Brush set MFJ-2012DS	1
Brush kit 5702643	1
Solenoid assembly 8737877	3
Solenoid kit 71220	3

Drive, brush, and solenoid samples shall be assembled in compatible starters, which shall be provided by the contractor. Parts shall be representative of units proposed to be furnished to the Government. Qualification testing shall be conducted under Government surveillance by the contractor, or by an authorized testing facility at a site approved by the Government.

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Qualification inspection shall consist of inspection as specified in 4.8.1; and a comparison examination of material, dimensions, and finish with applicable drawings or MS standards, or with another part having the same PIN that has already been qualified. After comparison examination, parts shall be assembled in a starter assembly, examined for the defects specified in table II, and tested as specified in table III. The applicable tests specified in tables IV, V, or VI shall then be performed for the repair part being qualified.

TABLE IV. Drive assemblies 11602684, 11667957, and 11667959.

Sample	Requirement	Test
A	3.6.2.4	4.8.18.4.2 Extreme temperature
B	3.6.1	4.8.17 Life test
C	3.4.5	4.8.9 Loaded static torque
D	3.4.6	4.8.10 Tooth abutment load
E	3.4.7	4.8.11 Overrunning torque

TABLE V. Brush assembly 7338603, brush assembly kit 71035, brush set MFJ-2012DS, and brush kit 5702643.

Sample 1/	Requirement	Test
A	3.4.1	4.8.3 Cranking torque and speed, 4.8.4 Slipping torque, 4.8.5 Stalling torque, or 4.8.6 Continuous load, whichever is applicable.
	3.6.2	4.8.18.1 Endurance, type I or 4.8.18.2.1 Endurance, type II, duty-light, classes 1 and 3.
	3.4.1	4.8.3 Cranking torque and speed, 4.8.4 Slipping torque, 4.8.5 Stalling torque, or 4.8.6 Continuous load, whichever is applicable.

1/ For brush assembly 7338603, eight samples shall be tested.

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TABLE VI. Solenoids 8737877 and 71220.

Sample	Requirement	Test
A	3.4.1 3.6.2	4.8.3 Cranking torque and speed 4.8.18.2 Endurance (duty-light)
B	3.4.1 3.5.1 3.4.4 3.5.3	4.8.3 Cranking torque and speed 4.8.12 Waterproofness 4.8.8 Dielectric strength 4.8.14 Fungus resistance (solenoid cover open during exposure)
C	3.4.1 3.6.2	4.8.3 Cranking torque and speed 4.8.18.3 Endurance (heavy-duty)

4.4.2 Retention of qualification. Certification shall be requested every 2 years from each manufacturer listed on the QPL to retain listing on the QPL. This certification shall be forwarded to the preparing activity and shall be signed by a responsible official of management, attesting that the listed product still meets the requirements of the current issue of the specification, is available from the listed plant, and can be produced under the same conditions as originally qualified; that is, same process, materials, construction, design, and manufacturers part number or designation. Failure to provide certification shall be cause for removal from the QPL.

4.4.3 Failure. Failure of a qualification sample to pass any of the examinations or tests specified herein may be cause for the Government to refuse to conduct additional testing until the faults revealed by the tests have been corrected.

4.5 QCI.

4.5.1 Sampling.

4.5.1.1 Lot formation. Unless otherwise specified (see 6.2), a lot shall consist of all the starters of one PIN, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

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4.5.1.3 Sampling for test. Samples for tests shall be selected in accordance with level S-3 of MIL-STD-105 from each lot that has passed the inspection specified in 4.5.2.2.

4.5.2 Examination.

4.5.2.1 Acceptable quality level (AQL). Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following AQL's:

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as specified in table II.

4.5.3 Tests. Samples selected in accordance with 4.5.1.3 shall be subjected to the tests specified in table VII, in the order specified, and shall conform to an AQL of 6.5 on the basis of percent defective.

TABLE VII. Order of quality conformance testing (starters).

Paragraph	Test
4.8.3	Cranking torque and speed
4.8.12.2	Waterproofness

4.6 Control tests. Sample starters shall be selected at the rate of 1 starter for each 250 produced, except that not more than 3 nor less than 1 shall be selected in any 30-day period. Control test samples shall be identified by production period, and by lot. Samples shall be examined as specified in 4.5.2 and tested as specified in tables VIII through X in the order listed. Acceptance of the lot, or the production, for that 30-day period, shall be held in abeyance until successful completion of control testing.

4.6.1 Failure. Failure of any sample to pass any of the specified control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

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TABLE VIII. Order of control testing (starters).

Paragraph	Test
4.8.3	Cranking torque and speed
4.8.12.1	Waterproofness
4.8.8	Dielectric strength
4.8.4	Slipping torque (type I)
4.8.6	Continuous load
4.8.7	Overspeed capability

TABLE IX. Order of control testing (drives).

Paragraph	Test
4.8.3	Cranking torque and speed
4.8.17.1	Short term drive test
4.8.11	Overrunning torque

TABLE X. Order of control testing (solenoids and brushes).

Paragraph	Test
4.8.3	Cranking torque and speed
4.8.18.2.1	Endurance [type II, duty-light, classes 1 and 3 (except 1,000 cycles of operation required at 77 + 15°F]
4.8.18.1	Endurance [type I, (except 100 cycles of operation required at 700 lb-ft)]

4.7 Inspection of packaging. The preservation, packaging, packing, and marking shall be inspected to determine conformance to 5.1 and 5.2 (see 6.2).

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 through 3.2.1.1, 3.3.1 through 3.3.3, 3.3.5 through 3.3.8, 3.3.10, and 3.7 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements.

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Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Examination. Conformance to 3.3, 3.3.4, 3.3.9, 3.7, 3.8, and 3.9 shall be determined by examination.

4.8.3 Cranking torque and speed. To determine conformance to 3.4.1, the starter shall be mounted on a test stand to simulate actual mounting in use. Recommended SAE practice for ring gear or starter-driven gear dimensions and pitch circle clearances shall be used. Torque shall be measured ahead of the gear train. A continuous direct load shall be applied for not longer than 5 seconds. Readings from not less than five points of measurement taken at amperage readings corresponding to respective horsepower points shall be taken. When necessary, additional readings may be taken to determine conformance to the applicable curve. Between readings, not less than 5 minutes cooling time to permit the frame temperature to return to room temperature shall be allowed. Alternate methods for accomplishing this test may be used provided they have been approved by the Government.

4.8.3.1 Type I. The starter shall be tested to determine conformance to 3.4.1.1.

4.8.4 Slipping torque (type I). When mounted as specified in 4.8.3, and with prony brake locked, the starter shall be energized by setting the voltage low and increasing it slowly until the clutch slips. Breakaway torque (highest reading reached before slippage) shall not be greater than that specified on the applicable drawing or military standard. The clutch shall be slipped for 3 seconds after the breakaway point and the torque reading taken within the following 2 seconds to determine conformance to 3.4.1.1. Three successive engagements shall be made with an off time between engagements of not less than 1 minute.

4.8.5 Stalling torque (type II). When mounted as specified in 4.8.3, the stalling torque and voltage of the starter shall be measured at the applicable current as shown on the applicable drawing or military standard to determine conformance to 3.4.1. Readings shall be taken within 5 seconds. When stalling torques are obtained as part of 4.8.3 tests, separate stall tests are unnecessary. In lieu of stall testing for a quality conformance test, testing to determine conformance to a suitable point of the starter output curve, as shown on the applicable drawing or military standard, may be substituted.

4.8.6 Continuous load. The test shall be conducted with voltage applied to the motor terminals. The voltage shall be in accordance with the battery duty curve for the current drawn on the applicable performance curve (see 3.3.10). Upon completion of this test the starter shall be tested as specified in 4.8.3 to determine conformance to 3.4.1.

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4.8.6.1 Type I. The starter shall operate under load for a continuous period of 6 minutes. The load shall be not less than 400 lb-ft. The starter terminal voltage shall be 13.5 ± 0.5 V dc.

4.8.6.2 Type II. The starter shall be subjected to seven cycles of operation, each cycle consisting of 30 seconds of continuous operation, exerting a load of such magnitude that starter current will be not less than 80% of that current associated with maximum horsepower as shown on the applicable performance curve, followed by 2 minutes off time. Immediately after the seventh cycle, the starter shall be subjected to the test specified in 4.8.7.

4.8.7 Overspeed capability. With 24 V properly applied at the motor terminals, the starter shall be operated for 1 minute at no load. After the test, the starter shall be subjected to the test specified in 4.8.3 to determine conformance to 3.4.3.

4.8.8 Dielectric strength. The starter shall be subjected to the test specified in method 301 of MIL-STD-202. Applied voltage shall be 500 V root mean square (rms) at a frequency of 60 hertz. The test voltage source shall have a minimum rating of 0.5 kilovolt-ampere. After the test, the starter shall be subjected to the test specified in 4.8.3 to determine conformance to 3.4.4.

4.8.9 Loaded static torque. To determine conformance to 3.4.5, drives shall be tested according to the following procedure:

a. Install drive in starter. The drive shall be energized into a stationary ring gear which shall provide 350 lb-ft of torque at the drive. There shall be no hesitation and the drive shall engage positively at this load.
b. The duration of activation shall not exceed 5 seconds.
c. Repeat test at 1 minute intervals four additional times.

4.8.10 Tooth abutment load. To determine conformance to 3.4.6, an axial force shall be applied to a single pinion tooth of the drive to produce its full indexing position. Resulting force shall be measured.

4.8.11 Overrunning torque. To determine conformance to 3.4.7, positork drives shall be tested according to the following procedure:

a. Rotate positork drive pinion in overrunning direction by mechanical means (see figure 1).
b. Measure speed of the drive pinion at which pinion and clutch dentils separate. Separation of the dentils can be determined by change in sound.
c. Increase speed of the drive pinion to 15,000 rpm, and measure overrunning torque of the drive.

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4.8.12 Waterproofness, grade A.

4.8.12.1 Qualification and control tests. With the drive end sealed, the starter shall be subjected to the waterproofness test for type III, class 1 components as specified in MIL-STD-1184. Before submersion, the starter shall be operated at no load and reduced voltage (18 to 20 V) for 60 seconds to insure proper operation. While submerged, the starter shall also be operated three times in the same manner at suitable intervals. After the test, the starter shall be subjected to the test specified in 4.8.3 to determine conformance to 3.5.1.1.

4.8.12.2 Quality conformance test. With the drive end sealed, the starter shall be subjected to a no load run of 10 seconds at a suitable reduced voltage, prior to submersion for 1 minute in tap water. While submerged, an internal pressure of 6 pounds per square inch (psi) shall be applied to determine conformance to 3.5.1.2.

4.8.13 Corrosion resistance. With the drive end sealed, the starter shall be subjected to the salt spray test specified in method 101 of MIL-STD-202, except that the test duration shall be 96 hours. After which the exposure shall be continued with drive end exposed for 96 hours. After the test, the starter shall be subjected to the tests specified in 4.8.8, 4.8.12, and 4.8.3 through 4.8.6 to determine conformance to 3.5.2.

4.8.14 Fungus resistance. The starter shall be subjected to the test specified in class 2, method B of MIL-F-13927, except that the test duration shall be a continuous 90 days. The starter need not be disassembled for this test, but brush covers/bands shall be removed and solenoid covers loosened to expose solenoid contacts. Upon removal from the cabinet, examination shall be accomplished by means of removable plates or plugs, without disassembly and without disturbing any fungus growths that have developed. Upon completion of the examination and replacement of the access part covers, the starters shall be rotated for 5 minutes before being subjected to the tests specified in 4.8.8 and 4.8.3 through 4.8.6 to determine conformance to 3.5.3.

4.8.15 Vibration resistance. The starter shall be mounted to simulate actual mounting in use and subjected to the vibration test specified in method 201 of MIL-STD-202 for 1 hour in each direction of the three major axes of the starter. After the test, the starter shall be subjected to the test specified in 4.8.3 to determine conformance to 3.5.4.

4.8.16 Shock resistance. The starter shall be mounted to simulate actual mounting in use and subjected to the shock test specified in method 213, test condition K of MIL-STD-202. After the test, the starter shall be subjected to the test specified in 4.8.3 to determine conformance to 3.5.5.

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4.8.17 Life test (live engine test). To determine conformance to 3.6.1, the applicable drive or clutch assembly shall be installed on a compatible cranking motor. The cranking motor shall then be installed on a four cycle compression ignition (diesel) of at least 500 cubic inch displacement (300 cubic inches for two cycle diesel). The engine shall be set so that the drive's pinion speed shall be at least 16,000 rpm during overrun. The overrun shall last 0.5 to 0.7 seconds after the engine starts. The test sequence shall require 160 start attempts per hour with engine rotation to be stopped after a minimum of 5 seconds (cranking time will be limited to 2 seconds if the engine fails to start).

4.8.17.1 Short term drive test (live engine). This test shall be conducted under conditions specified in 4.8.17. The applicable drive or clutch shall be tested for a minimum of 1,000 cycles.

4.8.18 Endurance.

4.8.18.1 Type I. When mounted on a test stand in a manner simulating the actual engine mounting, the starter shall be subjected to 6,400 cycles of operation. Each cycle shall consist of 15 seconds under load, followed by 2.5 minutes off time. The loads shall be as follows:

Starter load (lb-ft)	Number of cycles
700	2,000
600	2,000
400	1,200
200	1,200

At the end of each load cycle, a suitable over-run device shall be energized to return the starter jaw to its normal de-energized position. The test shall be continuous except for the time necessary to change loads (if required). The load sequence is optional except that the loads shall be periodically varied through the entire range (No single load shall be maintained for over 16 hours). Input voltage at the starter shall be 24.0 ± 0.5 V and the ambient temperature shall be 77 ± 10°F. Cooling air may be applied to the starter during the test but the waterproofing cylinder shall not be removed. Brush lengths and starter efficiency shall be obtained at the beginning and at the end of the test to determine conformance to 3.6.2.2.

4.8.18.2 Type II, duty-light.

4.8.18.2.1 Classes 1 and 3. When mounted as specified in 4.8.3, the starter shall be subjected to 5,000 cycles of operation, with each cycle consisting of 5 seconds of operation under load followed by 1 minute off time. The load shall be the minimum draw current associated with 75% of the maximum horsepower as specified on the applicable drawing or military

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standard. The load shall be applied at the starter pinion through ring gear engagement means as in intended operation. An air velocity of approximately 1,500 linear feet per minute to force ambient air over and around the starter is allowable. The 5,000 cycles of operation shall be conducted at the following ambient air temperatures:

Number of cycles	Temperature
1,000	160 ± 5°F
1,000	-65 ± 5°F
3,000	77 ± 15°F

Class 3 starters shall be mechanically actuated for each cycle. Prior to operation at these temperatures, the starter shall be stabilized at the temperature for 24 hours. During the last 1,000 cycles the longitudinal axis of the starter shall be positioned at an angle with the starter pinion 45 degrees above and below the horizontal for 500 cycles each. After testing, the starter shall be subjected to the test specified in 4.8.3 to determine conformance to 3.4.3.

4.8.18.2.2 Class 2 (inertia drive). It is optional that at the conclusion of each load cycle (see 4.8.18.2.1), an over-running device may be actuated to return the starter to its normal de-energized position.

4.8.18.2.3 Brush wear (classes 1 and 2). To determine conformance to 3.6.2.3, the starter, with proposed brushes installed, shall be tested as specified in 4.8.18.2.1 or 4.8.18.2.2 (as applicable), except for the following:

a. 7,000 cycles at 77 ± 15°F.
b. Observe for commutation and sparking irregularity; angle operation at 45 degrees above and below the horizontal is not required.

4.8.18.3 Type II, duty-heavy. When mounted as specified in 4.8.3, the starter shall be subjected to 1,000 cycles of operation, with each cycle consisting of 30 seconds of operation under load followed by 5.5 minutes off time. The load shall be so adjusted that the starter produces 95% of the maximum horsepower shown on the applicable drawing or standard. The load shall be applied at the starter pinion through ring gear engagement means as in intended operation. An air velocity of approximately 1,500 linear feet per minute to force ambient air over and around the starter is allowable. During the last 200 cycles, the longitudinal axis of the starter shall be positioned at an angle with the starter pinion 45 degrees above and below the horizontal for 100 cycles each. After testing, the starter shall be subjected to the test specified in 4.8.3 to determine conformance to 3.4.3.

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4.8.18.4 Extreme temperature.

4.8.18.4.1 Type II, duty-heavy. To determine conformance to 3.6.2.4, the starter shall be mounted as specified in 4.8.3. The starter shall be subjected to 400 cycles of operation, with each cycle consisting of 30 seconds of operation under load followed by 5.5 minutes off time. The load shall be so adjusted that the starter produces 95% of the maximum horsepower at the minimum current shown on the applicable drawing or standard. The load shall be applied at the starter pinion through ring gear engagement means as in intended operation. An air velocity of approximately 1,500 linear feet per minute to force ambient air over and around the starter is allowable. The 400 cycles of operation shall consist of 200 cycles at each of the following ambient air temperatures:

a. -65 ± 5°F.
b. 160 ± 5°F.

After testing, the starter shall be subjected to the test specified in 4.8.3 to determine conformance to 3.4.3.

4.8.18.4.2 Drive assemblies. To determine conformance to 3.6.2.4, the starter shall be mounted as specified in 4.8.3. The starter shall be subjected to 400 cycles of operation, with each cycle consisting of 30 seconds of operation under load followed by 5.5 minutes off time. The load shall be so adjusted that the starter produces 95% of the maximum horsepower at the minimum current shown on the applicable drawing or standard. The load shall be applied at the starter pinion through ring gear engagement means as in intended operation. An air velocity of approximately 1,500 linear feet per minute to force ambient air over and around the starter is allowable. The 400 cycles of operation shall consist of 200 cycles at each of the following ambient air temperatures:

a. -65 ± 5°F.
b. 160 ± 5°F.

After testing, the starter shall be subjected to the test specified in 4.8.3 to determine conformance to 3.4.3.

5. PACKAGING

5.1 Preservation, packing, and marking. The desired level shall be as specified (see 4.7 and 6.2).

5.1.1 Level A preservation. Level A preservation shall be in accordance with MIL-STD-2073-1, code 3Q100EAGCDEE or as an alternate code 3Q100EAGMHEE.

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5.1.2 Level A packing. Level A packing shall be in accordance with MIL-STD-2073-1, packing code Q.

5.1.3 Level B preservation. Level B preservation shall be the same as level A and is therefore not applicable.

5.1.4 Level B packing. Level B packing shall be in accordance with MIL-STD-2073-1, packing code F.

5.1.5 Marking. All unit packs and shipping containers shall be marked in accordance with MIL-STD-129 in addition to any special marking required by the contract (see 6.2).

5.2 Special packaging instructions (SPI). As an alternate to 5.1. when specified by the acquisition activity; preservation, packaging, packing, and marking for the desired level of protection shall be in accordance with the applicable SPI specified by the contracting authority (see 4.7 and 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The starters covered by this specification are intended to be used for cranking nonaircraft internal-combustion engines utilizing a 24 V dc electrical system.

6.1.1 Type I. Type I starters are intended for direct engagement to an engine crankshaft.

6.1.2 Type II. Type II starters are intended for use on engines equipped with a ring gear on the flywheel and are usually designated as automotive type.

6.2 Acquisition documents. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, grade, class, and duty of unit required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
d. Rotation, if other than clockwise (see 3.3.5.3).
e. Applicable military standard or drawing (see 3.3.10.1).
f. Performance curve required (see 3.3.10.2).
g. Military PIN and Federal stock number (see 3.8.1 and 3.8.2).
h. If responsibility for inspection and the place of inspection is other than the contractors (see 4.1).
i. Lot size, if other than as specified (see 4.5.1.1).

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j. Categories of inspection of packaging, sample size for each category, approval of test results and disposition of defective items (see 4.7).
k. Selection of applicable level and packaging requirements (see 5.1 and 5.1.5, or 5.2).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in QPL No. 3785 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the QPL is the Commander, US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, Michigan 48397-5000 and information pertaining to qualification of products may be obtained from that activity.

6.3.1 Extension of qualification.

6.3.1.1 Rotation. Qualification will be extended to starters having opposite direction of location (see 3.3.5.3) than the sample tested provided the design and construction features are the same and the direction of rotation plainly marked (see 3.8).

6.4 Definitions.

6.4.1 Duty-heavy. Heavy-duty starters are starters that are tested for conformance to the current issue of the SAE heavy-duty battery curve.

6.4.2 Duty-light. Light-duty starters are starters that are tested for conformance to the current issue of the SAE light-duty battery curve.

6.4.3 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.4.4).

6.4.4 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

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6.5 Subject term (key word) listing.

Clutch
Crankshaft
Direct engagement
Enamel paint, black gloss
Flywheel ring gear
Internal-combustion engines
Peripheral engagement
Pinion
Recovered materials

6.6 Changes from previous issue. Marginal motations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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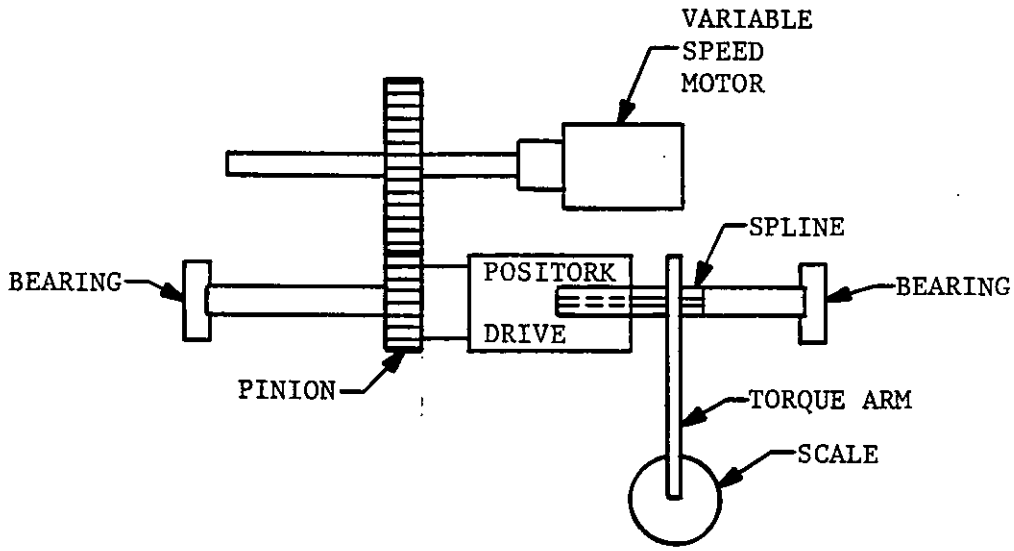


FIGURE 1. Mechanical means for conducting the overrunning torque test.

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Custodian:
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Review activity:
DLA - CS

User activity:
Army - ME

Preparing activity:
Army - AT

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-S-3785E(AT); Starters, Engine, Electrical: 24-Volt Direct-Current

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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